

Problem Specification

- **Task:** Matrix Multiplication $C = A \times B$
- **Matrix Size (N):** 32×32
- **Data Type:** FP32 (4 bytes per element)
- **Storage:** Row-major order

Task 1: Naive Triple Loop (ijk order)

In the naive implementation, we assume zero data reuse (every access result in a DRAM fetch).

- **Accesses to B per output element:** To compute one element $C[i][j]$, we perform a dot product of a row in A and a column in B. This requires 32 accesses to Matrix B.
- **Total element accesses (A and B):** Matrix A is accessed N times for every element in C (N^2 elements). Total = N^3 .
 - Matrix B is accessed N times for every element in C. Total = N^3 .
 - Total Accesses = $2 \times 32^3 = 65,536$ accesses
- **Total DRAM Traffic:**
 - We must account for reading A and B, and finally writing the result C back to DRAM.
 - Formula: $(N^3 + N^3 + N^2) \times 4$ bytes
 - Calculation: $(32,768 + 32,768) \times 4 = 65,536 \times 4 = 262,144$ Bytes

Task 2: Tiled Loop Analysis (Tile Size $T=8$)

By blocking the computation into 8×8 tiles, we utilize local SRAM (Shared Memory) to reuse data.

- **DRAM loads per tile:** To compute one $T \times T$ tile of C, we load a strip of tiles from A and B.
 - Number of tiles across the dimension = $N/T = 32/8 = 4$.
 - We load 4 tiles of A and 4 tiles of B from DRAM into local memory.
- **Total DRAM Traffic:**
 - With tiling (simplified model), each element of A and B is loaded from DRAM only once and reused on-chip.
 - Formula: $((N^2 \times N/T) + (N^2 \times N/T) + N^2) \times 4$ bytes
 - Calculation: $(1024 + 1024 + 1024) \times 4 = 3,072 \times 4 = 12,288$ Bytes

Task 3: Traffic Efficiency Ratio

- **Ratio:** Naive Traffic/Tiled Traffic = $262,144/8,192 = 32 = N$.
- **Justification:** The ratio equals N because tiling allows each element of A and B to be loaded once into shared memory and reused N times, reducing total DRAM traffic from $O(N^3)$ to $O(N^2)$.

Task 4: Execution Time and Classification

- **Assumed Specs:** Bandwidth = 320 GB/s, Peak Compute = 10 TFLOPS.
- **Total Operations:** $2 \times N^3 = 65,536$ FLOPs
- **Naive Execution Time:** $262,144 \text{ bytes} / (320 \times 10^9 \text{ bytes/s}) = 0.819 \text{ microseconds}$.
 - **Bound: Memory-Bound.** The low arithmetic intensity forces the kernel to be limited by the bandwidth roof.
- **Tiled Execution Time:** $12,288 \text{ bytes} / (320 \times 10^9 \text{ bytes/s}) = 0.038 \text{ microseconds}$.
 - **Bound: Compute-Bound** (or much closer to the ridge). By reducing memory traffic, the bottleneck shifts from the memory bus toward the actual processing units.